

CI/CD Integration

Series: AUTOM | Notebook: 7 of 8 | Created: January 2026

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Prerequisites

Before starting this notebook, ensure you have:

Requirement	Description
CI/CD Platform	GitHub Actions, GitLab CI, or Jenkins
Monaco or Terraform	One of the config-as-code tools
Git Repository	For storing configurations
API Token	Stored as CI/CD secret

Learning Objectives

By the end of this notebook, you will:

- Understand GitOps principles for Dynatrace
 - Know how to set up CI/CD pipelines for config deployment
 - Be able to implement pull request workflows
 - Handle multi-environment deployments
-

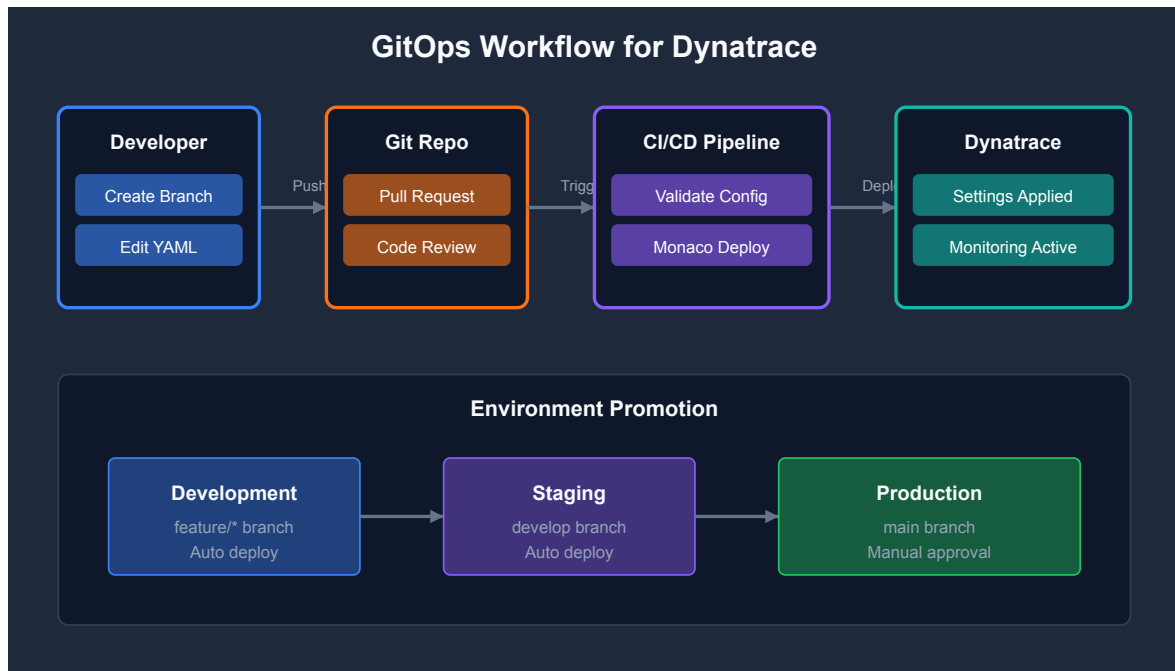
1. Introduction

CI/CD integration brings software development practices to Dynatrace configuration management. By storing configs in Git and deploying via pipelines, teams gain version control, review processes, and automated deployments.

Why CI/CD for Dynatrace?

Benefit	Description
Version Control	All changes tracked in Git history
Code Review	Pull requests for config changes
Automation	No manual deployments
Rollback	Revert to any previous state
Audit Trail	Who changed what, when

GitOps Workflow



2. GitOps Fundamentals

Repository Structure

```
dynatrace-config/  
├── .github/  
│   ├── workflows/  
│   │   ├── validate.yml  
│   │   └── deploy.yml  
├── manifest.yaml          # Monaco manifest  
├── environments/  
│   ├── dev.yaml  
│   ├── staging.yaml  
│   └── production.yaml  
├── projects/  
│   ├── alerting/  
│   ├── management-zones/  
│   └── synthetic/  
└── README.md
```

Branch Strategy

Branch	Purpose	Deploys To
main	Production configs	Production
develop	Integration branch	Staging
feature/*	New features	Dev (optional)

Environment Promotion



3. GitHub Actions

Validation Workflow

`.github/workflows/validate.yml:`

```

name: Validate Dynatrace Config

on:
  pull_request:
    branches: [main, develop]
    paths:
      - 'projects/**'
      - 'manifest.yaml'

jobs:
  validate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Validate Configuration
        run: monaco validate manifest.yaml

      - name: Dry Run Deploy
        env:
          DT_DEV_URL: ${ secrets.DT_DEV_URL }
          DT_DEV_TOKEN: ${ secrets.DT_DEV_TOKEN }
        run: |
          monaco deploy manifest.yaml --environment development --dry-run

```

Deployment Workflow

.github/workflows/deploy.yml:

```
name: Deploy Dynatrace Config

on:
  push:
    branches: [main]
    paths:
      - 'projects/**'
      - 'manifest.yaml'

jobs:
  deploy-staging:
    runs-on: ubuntu-latest
    environment: staging
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Deploy to Staging
        env:
          DT_STAGING_URL: ${ secrets.DT_STAGING_URL }
          DT_STAGING_TOKEN: ${ secrets.DT_STAGING_TOKEN }
        run: |
          monaco deploy manifest.yaml --environment staging

  deploy-production:
    needs: deploy-staging
    runs-on: ubuntu-latest
    environment: production
    steps:
      - uses: actions/checkout@v4

      - name: Install Monaco
        run: |
          curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/latest/download/monaco-linux-amd64 -o monaco
          chmod +x monaco
          sudo mv monaco /usr/local/bin/

      - name: Deploy to Production
        env:
          DT_PROD_URL: ${ secrets.DT_PROD_URL }
```

```
DT_PROD_TOKEN: ${ secrets.DT_PROD_TOKEN }  
run: |  
    monaco deploy manifest.yaml --environment production
```

Terraform Workflow

For Terraform-based deployments:

```
name: Terraform Dynatrace

on:
  pull_request:
    branches: [main]
  push:
    branches: [main]

jobs:
  terraform:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Setup Terraform
        uses: hashicorp/setup-terraform@v3
        with:
          terraform_version: 1.6.0

      - name: Terraform Init
        run: terraform init
        env:
          TF_VAR_dynatrace_url: ${ secrets.DT_URL }
          TF_VAR_dynatrace_token: ${ secrets.DT_TOKEN }

      - name: Terraform Plan
        if: github.event_name == 'pull_request'
        run: terraform plan -no-color
        env:
          TF_VAR_dynatrace_url: ${ secrets.DT_URL }
          TF_VAR_dynatrace_token: ${ secrets.DT_TOKEN }

      - name: Terraform Apply
        if: github.ref == 'refs/heads/main' && github.event_name ==
'push'
        run: terraform apply -auto-approve
        env:
          TF_VAR_dynatrace_url: ${ secrets.DT_URL }
          TF_VAR_dynatrace_token: ${ secrets.DT_TOKEN }
```

4. GitLab CI/CD

Pipeline Configuration

.gitlab-ci.yml:

```
stages:
  - validate
  - deploy-staging
  - deploy-production

variables:
  MONACO_VERSION: "2.0.0"

.monaco-setup: &monaco-setup
  before_script:
    - curl -L https://github.com/Dynatrace/dynatrace-configuration-as-code/releases/download/v${MONACO_VERSION}/monaco-linux-amd64 -o monaco
    - chmod +x monaco
    - mv monaco /usr/local/bin/

validate:
  stage: validate
  <<: *monaco-setup
  script:
    - monaco validate manifest.yaml
    - monaco deploy manifest.yaml --environment development --dry-run
  rules:
    - if: $CI_PIPELINE_SOURCE == "merge_request_event"

deploy-staging:
  stage: deploy-staging
  <<: *monaco-setup
  script:
    - monaco deploy manifest.yaml --environment staging
  environment:
    name: staging
  rules:
    - if: $CI_COMMIT_BRANCH == "main"

deploy-production:
  stage: deploy-production
  <<: *monaco-setup
  script:
    - monaco deploy manifest.yaml --environment production
  environment:
    name: production
  rules:
    - if: $CI_COMMIT_BRANCH == "main"
      when: manual
```

Variables Configuration

In GitLab: **Settings** → **CI/CD** → **Variables**

Variable	Protected	Masked
DT_DEV_URL	No	No
DT_DEV_TOKEN	No	Yes
DT_STAGING_URL	No	No
DT_STAGING_TOKEN	Yes	Yes
DT_PROD_URL	Yes	No
DT_PROD_TOKEN	Yes	Yes

5. ArgoCD Integration

For Kubernetes-native GitOps, use ArgoCD with Dynatrace configs.

ArgoCD Application for Monaco Configs

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-config
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/org/dynatrace-config.git
    targetRevision: HEAD
    path: projects
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

ArgoCD for Dynatrace Operator (DynaKube)

Deploy and manage the Dynatrace Operator via ArgoCD:

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace-operator
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/org/k8s-platform.git
    targetRevision: HEAD
    path: dynatrace
  destination:
    server: https://kubernetes.default.svc
    namespace: dynatrace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
    syncOptions:
      - CreateNamespace=true
```

Repository structure for Dynatrace Operator:

```
dynatrace/
├─ kustomization.yaml
├─ namespace.yaml
├─ operator.yaml           # Dynatrace Operator deployment
├─ dynakube.yaml          # DynaKube custom resource
├─ secrets/
└─ dynakube-secret.yaml   # External Secrets reference
```

External Secrets for Token Management

Use External Secrets Operator to sync tokens from secret stores:

```

apiVersion: external-secrets.io/v1beta1
kind: ExternalSecret
metadata:
  name: dynakube-tokens
  namespace: dynatrace
spec:
  refreshInterval: 1h
  secretStoreRef:
    name: aws-secrets-manager
    kind: ClusterSecretStore
  target:
    name: dynakube
    creationPolicy: Owner
  data:
    - secretKey: apiToken
      remoteRef:
        key: dynatrace/api-token
    - secretKey: dataIngestToken
      remoteRef:
        key: dynatrace/data-ingest-token

```

Using Config Management Plugins

argocd-cm ConfigMap:

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: argocd-cm
  namespace: argocd
data:
  configManagementPlugins: |
    - name: monaco
      generate:
        command: ["sh", "-c"]
        args: ["monaco deploy manifest.yaml --environment $ARGOCD_ENV_ENVIRONMENT"]

```

6. FluxCD Integration

FluxCD provides an alternative GitOps approach with a pull-based reconciliation model.

FluxCD vs ArgoCD

Feature	FluxCD	ArgoCD
Architecture	Controller-based	Server + UI
UI	Minimal (Weave GitOps)	Built-in web UI
Multi-tenancy	Namespace-based	Project-based
Helm Support	Native HelmRelease	Application CRD
Image Automation	Built-in	Requires Argo CD Image Updater

FluxCD for Dynatrace Operator

GitRepository source:

```
apiVersion: source.toolkit.fluxcd.io/v1
kind: GitRepository
metadata:
  name: dynatrace-config
  namespace: flux-system
spec:
  interval: 1m
  url: https://github.com/org/k8s-platform
  ref:
    branch: main
  secretRef:
    name: github-token
```

Kustomization for Dynatrace:

```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace
  namespace: flux-system
spec:
  interval: 10m
  targetNamespace: dynatrace
  sourceRef:
    kind: GitRepository
    name: dynatrace-config
  path: ./dynatrace
  prune: true
  healthChecks:
    - apiVersion: apps/v1
      kind: Deployment
      name: dynatrace-operator
      namespace: dynatrace
```

FluxCD HelmRelease for Dynatrace Operator

Deploy via Helm with FluxCD:

```
apiVersion: source.toolkit.fluxcd.io/v1beta2
kind: HelmRepository
metadata:
  name: dynatrace
  namespace: flux-system
spec:
  interval: 1h
  url: https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable

---
apiVersion: helm.toolkit.fluxcd.io/v2beta1
kind: HelmRelease
metadata:
  name: dynatrace-operator
  namespace: dynatrace
spec:
  interval: 5m
  chart:
    spec:
      chart: dynatrace-operator
      version: ">=1.0.0"
      sourceRef:
        kind: HelmRepository
        name: dynatrace
        namespace: flux-system
  values:
    installCRD: true
```

SOPS for Secret Management with FluxCD

Use SOPS to encrypt secrets in Git:


```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace-secrets
  namespace: flux-system
spec:
  interval: 10m
  sourceRef:
    kind: GitRepository
    name: dynatrace-config
  path: ./dynatrace/secrets
  prune: true
  decryption:
    provider: sops
    secretRef:
      name: sops-age
```

7. Dynatrace Operator GitOps Patterns

When deploying the Dynatrace Operator via GitOps, follow these patterns for production environments.

Important: Use `apiVersion: dynatrace.com/v1beta5` for Dynatrace Operator 1.7.0+. Earlier versions (v1beta1, v1beta2) are deprecated and no longer supported.

Multi-Cluster Deployment Pattern

For organizations with multiple Kubernetes clusters:

```

platform-gitops/
├─ clusters/
│  ├─ production-east/
│  │  └─ dynatrace/
│  │     ├─ kustomization.yaml
│  │     └─ dynakube-patch.yaml      # Cluster-specific overrides
│  ├─ production-west/
│  │  └─ dynatrace/
│  │     ├─ kustomization.yaml
│  │     └─ dynakube-patch.yaml
│  └─ staging/
│     └─ dynatrace/
│        ├─ kustomization.yaml
│        └─ dynakube-patch.yaml
└─ base/
   └─ dynatrace/
      ├─ kustomization.yaml
      ├─ namespace.yaml
      ├─ operator.yaml
      └─ dynakube.yaml              # Base DynaKube config

```

Base kustomization.yaml:

```

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - namespace.yaml
  - https://github.com/Dynatrace/dynatrace-
operator/releases/download/v1.3.2/kubernetes.yaml
  - dynakube.yaml

```

Cluster overlay (production-east):

```

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - ../../base/dynatrace
patchesStrategicMerge:
  - dynakube-patch.yaml

```

Cluster-specific patch:

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  oneAgent:
    cloudNativeFullStack:
      args:
        - --set-host-group=production-east
```

Multi-Tenant Observability Pattern

For clusters serving multiple teams with different Dynatrace tenants:

```
# Team A – uses tenant-a.live.dynatrace.com
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: team-a-dynakube
  namespace: dynatrace
spec:
  apiUrl: https://tenant-a.live.dynatrace.com/api
  tokens: team-a-tokens
  oneAgent:
    cloudNativeFullStack:
      namespaceSelector:
        matchLabels:
          dynatrace-tenant: team-a

---
# Team B – uses tenant-b.live.dynatrace.com
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: team-b-dynakube
  namespace: dynatrace
spec:
  apiUrl: https://tenant-b.live.dynatrace.com/api
  tokens: team-b-tokens
  oneAgent:
    cloudNativeFullStack:
      namespaceSelector:
        matchLabels:
          dynatrace-tenant: team-b
```

GitOps Health Checks

Configure sync health checks to verify Dynatrace deployment:

ArgoCD health check:

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: dynatrace
spec:
  # ... source config ...
  ignoreDifferences:
    - group: dynatrace.com
      kind: DynaKube
      jsonPointers:
        - /status # Ignore status field changes
```

FluxCD health check:

```
apiVersion: kustomize.toolkit.fluxcd.io/v1
kind: Kustomization
metadata:
  name: dynatrace
spec:
  healthChecks:
    - apiVersion: apps/v1
      kind: Deployment
      name: dynatrace-operator
      namespace: dynatrace
    - apiVersion: apps/v1
      kind: DaemonSet
      name: dynakube-oneagent
      namespace: dynatrace
```

Sealed Secrets Alternative

For Kubernetes-native secret encryption:

```
apiVersion: bitnami.com/v1alpha1
kind: SealedSecret
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  encryptedData:
    apiToken: AgBy8BYo...encrypted...
    dataIngestToken: AgCtr4s2...encrypted...
```

Generate sealed secrets:

```
# Seal the secret (use your actual tokens from Dynatrace)
kubectl create secret generic dynakube \
  --from-literal=apiToken=&lt;your-api-token&gt; \
  --from-literal=dataIngestToken=&lt;your-data-ingest-token&gt; \
  --dry-run=client -o yaml | \
  kubeseal --format yaml &gt; sealed-dynakube.yaml
```

8. Best Practices

Security

Practice	Description
Masked secrets	Always mask API tokens in CI/CD
Protected branches	Require reviews for main branch
Environment protection	Manual approval for production
Token rotation	Rotate tokens regularly
Least privilege	Minimal token scopes
External Secrets	Use ESO, SOPS, or Sealed Secrets for K8s

Workflow Design

Practice	Description
Validate first	Always validate before deploy
Dry run PRs	Show what would change
Staged rollout	Dev → Staging → Production
Manual gates	Require approval for production
Rollback plan	Know how to revert changes
Health checks	Verify deployment success

GitOps Tool Selection

Use Case	Recommended Tool
Simple CI/CD	GitHub Actions / GitLab CI
K8s-native, UI preferred	ArgoCD
K8s-native, Helm-first	FluxCD
Multi-cluster enterprise	ArgoCD or FluxCD

Pull Request Template

`.github/PULL_REQUEST_TEMPLATE.md`:

```
## Dynatrace Configuration Change

### Description
<!-- What configuration is being changed? -->

### Type of Change
- [ ] New configuration
- [ ] Modification to existing config
- [ ] Deletion of config

### Environments Affected
- [ ] Development
- [ ] Staging
- [ ] Production

### Validation
- [ ] Monaco validate passed
- [ ] Dry run successful
- [ ] Tested in dev environment

### Rollback Plan
<!-- How to revert if issues arise? -->
```

Notification Integration

Add deployment notifications:

```
# GitHub Actions notification step
- name: Notify Slack
  if: always()
  uses: slackapi/slack-github-action@v1
  with:
    payload: |
      {
        "text": "Dynatrace config deployment: ${ job.status }",
        "blocks": [
          {
            "type": "section",
            "text": {
              "type": "mrkdn",
              "text": "*Deployment to ${ github.ref_name }*\nStatus: ${ job.status }\nCommit: ${ github.sha }"
            }
          }
        ]
      }
    env:
      SLACK_WEBHOOK_URL: ${ secrets.SLACK_WEBHOOK }
```

9. Next Steps

Deployment Event Tracking

Send deployment events to Dynatrace:


```
- name: Send Deployment Event
  run: |
    curl -X POST "${{ secrets.DT_URL }}/api/v2/events/ingest" \
      -H "Authorization: Api-Token ${{ secrets.DT_TOKEN }}" \
      -H "Content-Type: application/json" \
      -d '{
        "eventType": "CUSTOM_DEPLOYMENT",
        "title": "Dynatrace Config Deployment",
        "properties": {
          "source": "github-actions",
          "commit": "${{ github.sha }}",
          "branch": "${{ github.ref_name }}"
        }
      }'
```

Continue the Series

Next Notebook	Focus
AUTOM-08: Migration Automation	Bulk configuration transfer between tenants

Additional Resources

- [GitHub Actions Documentation](#)
- [GitLab CI/CD Documentation](#)
- [ArgoCD Documentation](#)
- [FluxCD Documentation](#)
- [Dynatrace Operator](#)
- [Monaco GitHub Repository](#)
- [External Secrets Operator](#)

Summary

In this notebook, you learned:

- GitOps fundamentals for Dynatrace configuration
- Setting up GitHub Actions and GitLab CI/CD pipelines

- ArgoCD integration with External Secrets for token management
- FluxCD with HelmRelease and SOPS for secret encryption
- Dynatrace Operator GitOps patterns for multi-cluster and multi-tenant environments
- Best practices for security and workflow design

Key Takeaway: CI/CD integration transforms Dynatrace config management into a software development workflow. For Kubernetes environments, GitOps tools like ArgoCD and FluxCD enable declarative management of both Dynatrace configuration and the Dynatrace Operator itself.

Continue to AUTOM-08: Migration Automation to learn bulk configuration transfer.

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.